

Several Functions

Look at this code:

```
def double(number):  
    return number * 2  
def square(number):  
    return number * number  
print("1: double\n2: square")  
choice = input("Choice: ")  
value = int(input("Number: "))  
if choice == "1":  
    print("Result:", double(value))  
else:  
    print("Result:", square(value))
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Rename `square(number)` to `triple(number)` and `return number * 3`. Change menu option 2 from "square" to "triple", then call `triple(value)` when [choice] is "2". Test "2" and 6; display "Result: 18".

Challenge 2

Add a function called `half(number)` that returns `number / 2`. Add "3: half" to the menu and use an `elif` condition to call `half(value)` when [choice] is "3". Add an `else` block that displays "Invalid choice." Test "3" and 5; display "Result: 2.5". Test "x" and 5; display "Invalid choice."

Challenge 3

Write `show_menu()` to display "1: area, 2: perimeter". Write `area(length, width)` to `return length * width`. Write `perimeter(length, width)` to `return 2 * length + 2 * width`. In the main program, call `show_menu()`, then ask for [choice], [length] and [width]. Display "Area:" when [choice] is "1", "Perimeter:" when [choice] is "2", and "Invalid choice." for any other value. Test "1", 4 and 3; display "Area: 12.0". Test "2", 4 and 3; display "Perimeter: 14.0". Test "x", 4 and 3; display "Invalid choice."

